AI-Assisted Web Portfolio Project Report

Design of Web-Based Systems - Mini Project

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1. Project Overview

The objective of this mini project was to create a professional online CV and portfolio webpage using HTML, CSS, and JavaScript while applying AI-assisted development to improve the structure, visual design, interactivity, and API integration. The website was based on Ahmed El Dessouky's CV content and redesigned into a modern, responsive, and interactive portfolio suitable for online deployment.

The final website contains sections for profile, education, experience, technical skills, projects, leadership activities, and contact information. It also includes JavaScript-based interactivity and a live API section connected to the GitHub REST API. The project was prepared as a static website so it can be deployed on GitHub Pages, Netlify, or any similar hosting service without requiring a backend server.

2. Features Implemented

- Professional landing page with profile image, name, title, quick highlights, and call-to-action buttons.
- Responsive navigation bar with a mobile menu for smaller screens.
- Dark/light mode toggle stored using localStorage so the selected theme remains active after refresh.
- About section toggle allowing users to switch between a full biography and a shorter summary.
- Interactive projects section with category filters, search box, result counter, and clickable project detail modal.
- Dynamic GitHub API integration that fetches public repositories by username and displays repository name, description, language, stars, and forks.
- Contact form with text fields, dropdown menu, radio buttons, checkbox, textarea, validation, and mailto email preparation.
- Download CV button linked to the included CV file in the assets folder.
- Scroll progress bar and reveal animations for a smoother user experience.
- Fallback messages and error handling for API failures, invalid GitHub usernames, or empty search results.

3. AI Tool Used

Tool Name	Platform / Version	Purpose
ChatGPT	GPT-5.5 Thinking	Assisted with planning the portfolio structure, generating initial HTML/CSS/JavaScript code, improving the interface, writing API handling logic, debugging functionality, and drafting the report.

4. Tasks Assisted by AI

Task	AI Contribution	Manual Modifications / Final Decisions
HTML Structure	Generated semantic page sections such as hero, about, education, experience, projects, API integration, leadership, and contact.	Content was adjusted to match the CV accurately, headings were organized, accessibility labels were added, and the downloadable CV link was included.
CSS Styling	Suggested a modern visual layout using cards, spacing, rounded corners, responsive grids, and a blue/white professional color palette.	The design was refined to improve visual hierarchy, mobile responsiveness, and consistency across dark and light modes.
JavaScript Interactivity	Generated functions for menu toggling, theme switching, project filtering, form validation, API fetching, and modal behavior.	Logic was revised to include error handling, saved theme preference, search/filter combination, keyboard accessibility, and safer input validation.
API Integration	Suggested using the GitHub REST API through fetch to load dynamic repository content.	Added username input, loading state, clear button, API failure message, empty result handling, and repository metadata display.

5. Functionality Testing

- Navigation links scroll correctly to their matching sections.
- Mobile menu opens and closes using the menu button, and closes after clicking a navigation link.
- Dark mode changes the interface colors and remains saved after refreshing the browser.
- About toggle correctly switches between the full and short profile descriptions.
- Project category filters work individually and together with the search input.
- Clicking a project opens a modal with project details; the modal closes using the close button, outside click, or keyboard escape action.
- GitHub API section successfully attempts to fetch repositories and shows a clear fallback message if the request fails.
- Contact form prevents invalid submission and only prepares an email after all required fields are valid.
- CV download button points to the included CV file inside the assets folder.
- The layout remains usable on desktop, tablet, and mobile screen widths.

6. Code Structure and Main Files

The project was organized into separate files to keep the code readable and easy to maintain. The HTML file controls the content and structure, the CSS file controls the design and responsiveness, and the JavaScript file controls the interactive behavior and API requests.

- **index.html:** Contains all portfolio sections, form fields, modal elements, and API section markup.
- **styles.css:** Contains layout rules, theme variables, responsive design, cards, buttons, and visual states.
- **script.js:** Contains DOM selection, event listeners, project filtering, modal handling, form validation, theme persistence, and GitHub API logic.
- **assets folder:** Contains the profile image and downloadable CV file.
- **report.pdf:** Documents the AI-assisted workflow, testing, reflections, and ethical considerations.

7. Challenges Encountered

One challenge was making sure the project did not only look like a static CV. To satisfy the interactive and API requirements, the project needed meaningful JavaScript features rather than simple decoration. Another challenge was making the GitHub API section reliable, since public API calls can fail because of internet issues, username errors, or rate limits. To address this, the code includes loading messages, error messages, and empty-state handling.

A third challenge was ensuring the design remained responsive because the CV content contains many sections and could easily become crowded on mobile screens. The solution was to use responsive grids, collapsible navigation, smaller spacing on narrow screens, and readable card layouts. The final challenge was making sure AI-generated code was not accepted without review. Each feature was checked manually and adjusted to match the intended behavior.

8. Reflections on AI Usage

What worked well: AI was useful for quickly generating the first version of the webpage structure and suggesting modern CSS styling patterns. It also helped create JavaScript functions for filtering, form validation, and API fetching, which saved time during development.

What needed manual correction: AI-generated code still required review. Some features needed stronger validation, clearer fallback messages, better accessibility, and more polished spacing. The content also needed to be checked against the original CV to avoid inaccurate wording.

Productivity impact: AI improved productivity by reducing the time needed to create the initial structure. Instead of starting from a blank page, the development process became more focused on improving the interface, testing behavior, and correcting details.

Learning impact: Using AI did not remove the need to understand the code. We still needed to understand HTML structure, CSS responsive design, JavaScript DOM manipulation, fetch API behavior, and validation logic to make sure the final website worked correctly.

Future use: We would use AI coding tools again for brainstorming, drafting boilerplate, debugging, and improving design ideas, but would continue to manually test and review the code before submitting or publishing it.

9. Ethical and Practical Considerations

Using AI in development creates several ethical and practical responsibilities. First, AI-generated code should not be accepted blindly because it may contain errors, accessibility issues, security weaknesses, or incomplete logic. Second, academic honesty requires clearly documenting which parts were assisted by AI, especially in coursework. Third, relying too heavily on AI can weaken learning if students do not understand the generated code. Finally, developers should be careful with personal data and should not paste private credentials, API keys, or sensitive information into AI tools.

In this project, AI was used as an assistant rather than a replacement for development. The final code was reviewed, customized, and tested so that the webpage met the project requirements and represented the CV content accurately.

10. Limitations and Future Improvements

The project currently works as a static frontend website. This is suitable for GitHub Pages, but it means that contact form submissions are prepared using mailto rather than being stored in a backend database. In the future, the project could be improved by adding a real backend contact endpoint, analytics, a project admin panel, and more advanced API integrations such as LinkedIn or GitHub contribution statistics.

Another possible improvement is to add more accessibility testing, including full keyboard navigation checks, color contrast verification, and screen reader testing. The design could also be expanded with downloadable PDF CV support, more project screenshots, and a live deployment link once the site is hosted online.

11. File Structure

The submission ZIP file follows the required project structure:

- index.html
- styles.css
- script.js
- assets/profile.png
- assets/Ahmed_El_Dessouky_CV.docx
- link.txt
- report.pdf
- README.txt

12. Online Deployment

The project is ready to be uploaded to GitHub Pages, Netlify, or any static hosting service. After deployment, the public URL should be copied into link.txt. The website uses only static files, so it does not require a backend server. The GitHub API integration runs directly from the browser using JavaScript fetch.

13. Conclusion

This project helped transform a traditional CV into a professional web-based portfolio with interactive features and dynamic content. AI assistance supported the development process by speeding up coding and improving design ideas, but the final result still required careful editing, testing, and understanding. The completed website demonstrates HTML structure, CSS styling, JavaScript interactivity, API integration, responsive design, and clear documentation of AI-assisted development.